

## FORT STOCKTON, TX, USA

WMO: 722618

Lat: 30.912N

Lon: 102.917W

Elev: 918

StdP: 90.78

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 23091

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.0       | -2.1 | -17.6                         | 0.9 | 11.6 | -15.9 | 1.0 | 12.6 | 12.9                  | 16.1 | 11.5 | 15.1 | 3.6                   | 40   | 0.556 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.4                   | 38.6            | 18.7 | 37.4 | 19.1 | 36.2 | 19.1 | 22.5                | 31.2 | 21.9 | 30.8 | 21.6 | 30.3 | 4.6                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.5                            | 17.0 | 24.7 | 19.8 | 16.2 | 24.3 | 19.1 | 15.5 | 24.4 | 70.9          | 31.0 | 68.7 | 30.8 | 67.0 | 30.5 | 25.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.5          | 10.3 | 9.0 | DB                         | -7.7 | 41.2               | 2.2 | 1.3                                                | -9.3  | 42.1       | -10.6 | 42.9       | -11.8 | 43.6       | -13.4 | 44.5 | (4) |
| (5)               |      |     | WB                         | -8.9 | 23.5               | 2.2 | 0.8                                                | -10.4 | 24.1       | -11.7 | 24.6       | -12.9 | 25.0       | -14.5 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 19.7        | 9.7   | 12.1  | 15.9  | 20.5  | 24.6  | 28.1  | 28.2  | 28.1  | 24.7  | 20.1  | 14.1  | 9.9   | (6)  |  |
|                                                                        | DBStd       | 8.09        | 5.46  | 5.68  | 5.04  | 4.43  | 3.98  | 2.66  | 2.25  | 2.22  | 3.12  | 4.42  | 5.11  | 5.21  | (7)  |  |
|                                                                        | HDD10.0     | 217         | 73    | 40    | 12    | 1     | 0     | 0     | 0     | 0     | 0     | 2     | 22    | 67    | (8)  |  |
|                                                                        | HDD18.3     | 1031        | 269   | 182   | 104   | 30    | 6     | 0     | 0     | 0     | 2     | 34    | 141   | 264   | (9)  |  |
|                                                                        | CDD10.0     | 3757        | 64    | 98    | 196   | 316   | 453   | 544   | 563   | 559   | 442   | 314   | 144   | 63    | (10) |  |
|                                                                        | CDD18.3     | 1529        | 2     | 7     | 29    | 94    | 201   | 294   | 305   | 301   | 194   | 87    | 13    | 2     | (11) |  |
|                                                                        | CDH23.3     | 18354       | 56    | 150   | 468   | 1191  | 2441  | 3676  | 3716  | 3616  | 1937  | 863   | 194   | 48    | (12) |  |
|                                                                        | CDH26.7     | 9490        | 6     | 36    | 154   | 543   | 1322  | 2099  | 2047  | 1981  | 917   | 337   | 44    | 4     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.8         | 4.5   | 4.9   | 5.2   | 5.5   | 5.4   | 5.5   | 4.9   | 4.4   | 4.3   | 4.5   | 4.4   | 4.4   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 347         | 13    | 14    | 11    | 19    | 37    | 42    | 40    | 45    | 63    | 36    | 19    | 14    | (15) |  |
|                                                                        | PrecMax     | 638         | 43    | 65    | 43    | 116   | 148   | 106   | 110   | 140   | 256   | 212   | 92    | 73    | (16) |  |
|                                                                        | PrecMin     | 176         | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 117         | 12    | 17    | 11    | 24    | 32    | 31    | 31    | 30    | 55    | 42    | 23    | 16    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 27.5  | 30.3  | 32.5  | 35.9  | 39.0  | 41.2  | 39.0  | 39.0  | 37.4  | 33.9  | 30.6  | 27.3  | (19) |  |
|                                                                        |             | MCWB        | 11.6  | 12.6  | 13.4  | 15.0  | 17.2  | 18.1  | 20.2  | 19.9  | 18.8  | 17.5  | 14.6  | 11.8  | (20) |  |
|                                                                        | 2%          | DB          | 24.9  | 27.4  | 30.5  | 33.8  | 37.4  | 39.0  | 37.5  | 37.4  | 35.2  | 32.3  | 27.7  | 24.5  | (21) |  |
|                                                                        |             | MCWB        | 10.9  | 11.4  | 12.8  | 14.3  | 17.0  | 18.4  | 20.2  | 20.0  | 19.0  | 17.4  | 13.7  | 10.9  | (22) |  |
|                                                                        | 5%          | DB          | 22.3  | 25.0  | 28.1  | 32.3  | 35.8  | 37.5  | 36.3  | 36.2  | 33.5  | 30.7  | 25.9  | 22.0  | (23) |  |
|                                                                        |             | MCWB        | 9.9   | 10.8  | 12.6  | 14.2  | 16.1  | 18.8  | 20.2  | 20.1  | 19.0  | 16.8  | 13.0  | 10.2  | (24) |  |
|                                                                        | 10%         | DB          | 18.8  | 22.3  | 26.2  | 30.1  | 33.6  | 35.9  | 35.0  | 34.8  | 32.1  | 28.3  | 22.9  | 18.6  | (25) |  |
|                                                                        |             | MCWB        | 8.7   | 9.9   | 11.9  | 13.9  | 16.2  | 19.1  | 20.1  | 20.1  | 19.1  | 16.3  | 12.4  | 9.0   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 14.4  | 15.2  | 18.0  | 19.9  | 21.9  | 23.3  | 23.4  | 22.9  | 22.4  | 21.6  | 18.1  | 14.5  | (27) |  |
|                                                                        |             | MCDB        | 21.1  | 22.2  | 24.4  | 27.1  | 29.9  | 32.2  | 31.4  | 31.2  | 29.8  | 28.5  | 23.1  | 20.6  | (28) |  |
|                                                                        | 2%          | WB          | 12.3  | 13.5  | 16.4  | 18.7  | 20.8  | 22.4  | 22.4  | 22.2  | 21.7  | 20.5  | 16.2  | 12.5  | (29) |  |
|                                                                        |             | MCDB        | 20.5  | 21.4  | 22.2  | 25.3  | 28.0  | 31.4  | 31.3  | 31.2  | 29.3  | 26.7  | 21.5  | 20.2  | (30) |  |
|                                                                        | 5%          | WB          | 10.8  | 12.1  | 15.3  | 17.6  | 20.2  | 21.8  | 21.8  | 21.7  | 21.1  | 19.6  | 15.0  | 11.1  | (31) |  |
|                                                                        |             | MCDB        | 19.9  | 21.9  | 21.6  | 24.3  | 27.3  | 30.6  | 30.8  | 30.9  | 28.4  | 25.4  | 21.7  | 19.7  | (32) |  |
|                                                                        | 10%         | WB          | 9.5   | 10.8  | 13.9  | 16.4  | 19.4  | 21.2  | 21.3  | 21.2  | 20.6  | 18.7  | 13.8  | 9.7   | (33) |  |
|                                                                        |             | MCDB        | 18.5  | 21.1  | 21.7  | 24.4  | 26.9  | 29.9  | 30.3  | 30.4  | 27.5  | 24.4  | 21.2  | 18.2  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 13.6  | 14.4  | 15.0  | 15.4  | 14.6  | 13.7  | 12.4  | 12.6  | 12.2  | 13.2  | 13.4  | 13.3  | (35) |  |
|                                                                        |             | MCDBR       | 17.4  | 18.2  | 18.0  | 17.8  | 17.1  | 16.0  | 14.4  | 14.2  | 14.6  | 15.5  | 16.8  | 17.0  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.3   | 8.3   | 7.4   | 7.2   | 6.5   | 4.6   | 3.9   | 3.6   | 4.4   | 5.6   | 7.1   | 8.0   | (37) |  |
|                                                                        |             | MCWBR       | 15.5  | 16.2  | 15.1  | 15.1  | 14.5  | 13.5  | 12.1  | 12.2  | 11.7  | 12.1  | 14.1  | 15.2  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.261 | 0.269 | 0.289 | 0.324 | 0.344 | 0.370 | 0.389 | 0.365 | 0.347 | 0.302 | 0.283 | 0.263 | (40) |  |
|                                                                        |             | taud        | 2.606 | 2.580 | 2.514 | 2.408 | 2.399 | 2.387 | 2.375 | 2.448 | 2.453 | 2.580 | 2.557 | 2.599 | (41) |  |
|                                                                        | Ebn at Noon |             | 989   | 1006  | 999   | 966   | 942   | 912   | 893   | 914   | 923   | 954   | 951   | 969   | (42) |  |
|                                                                        |             | Edh at Noon | 78    | 89    | 102   | 118   | 120   | 121   | 122   | 112   | 107   | 88    | 82    | 75    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 3.59  | 4.48  | 5.72  | 6.86  | 7.26  | 7.46  | 7.06  | 6.58  | 5.60  | 4.69  | 3.78  | 3.27  | (44) |  |
|                                                                        | RadStd      |             | 0.34  | 0.44  | 0.36  | 0.43  | 0.44  | 0.34  | 0.39  | 0.34  | 0.36  | 0.48  | 0.37  | 0.28  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (14 neighbors) |  | N/A   | N/A     | N/A    | +0.48   | N/A   | N/A   | N/A         | N/A     | +153    | +110    |

Nomenclature: See separate page